on accuracy, while a lower threshold prioritizes efficiency.

Step 3: Confidence-Guided Multiagent Collaboration The collaboration process involves response generation and refinement. In our experiments, we set up an environment with two rounds, each consisting of three agents. To conduct multiagent collaboration, in round 1, we obtain responses r_2 and r_3 from additional agents $\mathcal{A}_2$ and $\mathcal{A}_3$, along with their respective confidence scores c_2 and c_3. These confidence scores are explicitly concatenated with each response to convey the agent's certainty level in subsequent rounds.

After all agents generate their responses in the first round, the second round begins. Each agent is given responses from the other two agents, excluding its own. Using this additional context, each agent refines its reasoning and generates an updated response. This allows each agent to leverage confident and compelling arguments in the updated response.

Step 4: Final Answer Generation The responses generated in Step 3 serve as ingredients for deriving the final answer. We explore two distinct strategies for final answer output: voting-based selection and judge-based generation. The voting-based approach determines the final answer by selecting the most frequent response among all agent-generated outputs. We design this majority voting to enhance robustness by leveraging consensus across multiple perspectives, effectively mitigating individual agent biases. The judge-based approach introduces an additional judging agent, which generates the final response based on all agent outputs. We induce a judging mechanism to update the final response, prioritizing well-supported and coherent arguments.

3.2 Threshold Selection Strategy

To select the optimal confidence threshold $\theta^* \in \Theta$ that jointly maximizes predictive accuracy and inference-time efficiency, we introduce a scoring function that incorporates two soft penalty terms for both objectives. To support high-performance threshold selection, we define a statistically grounded high-performance region for DOWN, derived by computing the one-sided 95% Wilson lower bound of the maximum observed accuracy. Threshold candidates whose accuracy lies within this region are considered statistically indistinguishable from the best-performing value, whereas those outside the region are penalized via a soft penalty term. Let $A_\theta = \frac{k}{N}$ and $S_\theta = \frac{m}{N}$ denote the accuracy and debate skip rate (i.e. efficiency) for a given threshold value θ in the candidate set Θ. Given the highest observed accuracy $A_\theta^{\max}$ obtained from $k^{\max}$ correct predictions, the Wilson lower bound $\tilde{A}_\theta^{\max}$ is computed as:

$$\tilde{A}_\theta^{\max} = \frac{2k^{\max} + z^2 - z\sqrt{z^2 + 4k^{\max}\left(1 - \frac{k^{\max}}{N}\right)}}{2(N + z^2)}, \tag{2}$$

where $z = \Phi^{-1}(0.95) = 1.645$. Thresholds with raw accuracy below $\tilde{A}_\theta^{\max}$ are subject to soft penalization.

In parallel, to ensure a balanced trade-off between computational efficiency and performance improvement through debate, we penalize thresholds whose skip rate S_θ falls outside an interval $[s_{\min}, 1 - s_{\min}]$. Here, $s_{\min}$ denotes the proportion symmetrically excluded from the lower and upper bounds of the unit interval $[0, 1]$. The total penalty term is defined as: $P_\theta = \max(0, \tilde{A}_\theta^{\max} - A_\theta) + \max(0, s_{\min} - S_\theta) + \max(0, S_\theta - (1 - s_{\min}))$.

With both penalty terms defined, we compute the overall utility score for each threshold candidate. To ensure that accuracy and efficiency contribute comparably to the final score, we normalize both metrics via min-max scaling:

$$\hat{A}_\theta = \frac{A_\theta - A_\theta^{\min}}{A_\theta^{\max} - A_\theta^{\min} + \epsilon}, \hat{S}_\theta = \frac{S_\theta - S_\theta^{\min}}{S_\theta^{\max} - S_\theta^{\min} + \epsilon}, \tag{3}$$

where $\epsilon \ll 1$ is a small constant added for numerical stability. We compute the final score as the sum of the normalized accuracy $\hat{A}_\theta$ and efficiency $\hat{S}_\theta$, with a penalty term P_θ scaled by a weight factor λ. The optimal threshold is then determined by maximizing this score: $\theta^* = \arg\max_{\theta \in \Theta}(\hat{A}_\theta + \hat{S}_\theta - \lambda \cdot P_\theta)$.

4 Experiments

4.1 Experimental Setup

Models. We evaluate two agent configurations: homogeneous and mixed. The homogeneous setup leverages a single model across all agents, experimenting with Llama-3.1 8B (`meta-llama/Llama-3.1-8B-Instruct`), Ministral 8B (`mistralai/Ministral-8B-Instruct-2410`), Qwen-2.5 72B (`Qwen/Qwen2.5-72B-Instruct`), Llama-3.3 70B (`meta-llama/Llama-3.3-70B-Instruct`), and GPT-4o-mini

Method	Llama-3.1 8B		Ministral 8B		GPT-4o-mini		Llama-3.3 70B		Qwen-2.5 72B	
	Acc.	AC	Acc.	AC	Acc.	AC	Acc.	AC	Acc.	AC
Single-CoT	42.95	1.00	51.06	1.00	55.75	1.00	56.33	1.00	57.80	1.00
Self-refine	39.46	6.00	36.90	6.00	54.29	6.00	53.67	6.00	58.47	6.00
Self-Consistency	44.70	9.00	48.24	9.00	55.88	9.00	**58.18**	9.00	58.29	9.00
MAD	32.39	3.09	28.67	3.01	43.23	3.02	51.22	3.00	49.13	3.04
Debate	44.56	6.00	48.54	6.00	57.32	6.00	57.28	6.00	58.69	6.00
DOWN-Vote	45.51	1.50	**53.71**	1.48	57.09	1.80	57.80	1.02	59.39	1.28
DOWN-Judge	**45.52**	1.60	**53.71**	1.57	**57.35**	1.96	57.80	1.03	**59.52**	1.34

Table 1: Comparison of accuracy (Acc.) and average agent calls (AC) across single-agent methods, multiagent debate systems, and our proposed approach on the MUSR dataset. Single CoT, Self-refine, and Self-consistency are single model-based approaches, while MAD, Debate, and DOWN are multiagent debate-based systems.

Method	Llama-3.1 8B		Ministral 8B		GPT-4o-mini		Llama-3.3 70B		Qwen-2.5 72B	
	Acc.	AC	Acc.	AC	Acc.	AC	Acc.	AC	Acc.	AC
Single-CoT	70.74	1.00	67.69	1.00	78.17	1.00	80.35	1.00	78.40	1.00
Self-refine	69.54	6.00	67.69	6.00	76.42	6.00	77.73	6.00	78.17	6.00
Self-Consistency	68.56	9.00	68.12	9.00	79.48	9.00	80.79	9.00	77.29	9.00
MAD	44.54	4.66	57.64	3.73	70.31	3.38	79.04	3.07	73.80	3.24
Debate	70.08	6.00	**70.74**	6.00	79.04	6.00	80.35	6.00	**79.91**	6.00
DOWN-Vote	**71.18**	2.53	68.12	3.16	**80.79**	1.92	**82.53**	1.07	77.73	2.64
DOWN-Judge	69.87	2.83	68.12	3.59	79.91	2.10	**82.53**	1.08	77.73	2.97

Table 2: Comparison of accuracy (Acc.) and average agent calls (AC) across single-agent methods, multiagent debate systems, and our proposed approach on the StrategyQA dataset

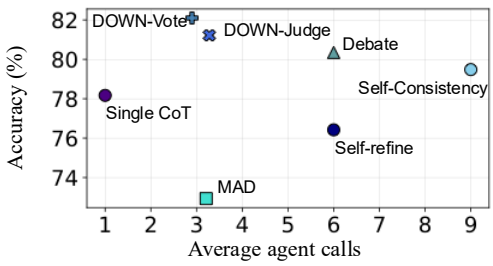
(`gpt-4o-mini`). The mixed configuration utilizes Qwen-2.5 72B, Llama-3.3 70B, and GPT-4o-mini, with the order of model selection randomized for each query. Each debate round consists of three agents, and we conduct a two-round debate where responses are generated in the first round and refined in the second.

Tasks. For evaluation, we utilize benchmarks specifically designed to assess reasoning capabilities. MUSR (Sprague et al., 2024) evaluates multi-step soft reasoning over free-text narratives, offering a more complex yet realistic reasoning challenge compared to synthetic benchmarks. Assessments are performed on 756 samples drawn from the murder mysteries, object placements, and team allocation subsets. StrategyQA (Geva et al., 2021) requires implicit reasoning, where inference steps must be strategically derived rather than explicitly provided. Evaluations are conducted on the development set comprising 229 samples.

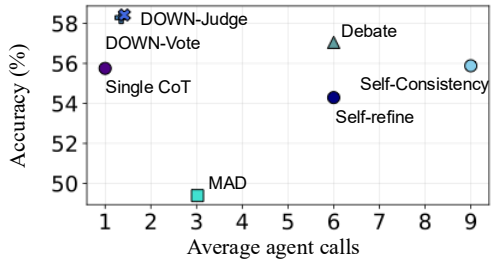
Baselines. We compare our approach against multiple established reasoning frameworks, along with a single LLM CoT (Kojima et al., 2022) baseline. (1) **Self-refine** (Madaan et al., 2023): Generating an initial response and iteratively refining its output through self-feedback. We design an environment composed of two rounds, with each round comprising three agents. (2) **Self-Consistency** (Wang et al., 2023): Sampling diverse reasoning paths and deriving the final answer by selecting the most consistent one through majority voting. We sample nine responses for each query. (3) **MAD** (Liang et al., 2024): Conducting a debate between two agents with opposing perspectives while a moderator selects the most plausible solution or continues the debate if needed. (4) **Debate** (Du et al., 2024): Engaging agents in a structured debate, iteratively refining their responses by incorporating insights from previous exchanges. We design an environment composed of two rounds, with each round comprising three agents.

4.2 Implementation Details

We set the temperature to 0.0 to ensure deterministic response generation and limit the maximum sequence length to 512 tokens. The experimental setup maintains consistent configurations across all models. For the θ^* selection, we define the candidate set as $\Theta = \{0.7, 0.8, 0.9\}$, starting from a threshold that yields a debate skip rate S_θ close to 1. The symmetric margin $s_{\min}$ and the penalty



(1) MUSR dataset



(2) StrategyQA dataset

Figure 3: Comparison of multiagent debate system performance in a mixed-model configuration. The configuration includes Llama3.3-70B, Qwen-2.5 72B, and GPT-4o-mini, with the model order randomized for each query. For single model-based approaches, we present the results of GPT-4o-mini.

scaling factor λ are set to 0.1 and 15. For the evaluation metrics, we use accuracy as the metric, while efficiency is assessed based on the average number of agent calls or debate skip rate. The experiments are conducted using four 48GB A6000 GPUs. All prompts used in the experiments are provided in Table 6.

5 Results and Analysis

5.1 Main Results

Homogeneous-Model Configuration Table 1 presents the results on the MUSR dataset. Unlike prior approaches that require three to nine agent calls per query, our framework achieves comparable or superior performance with only 1.4 calls on average, representing over a sixfold improvement in efficiency. The results in Table 2 on the StrategyQA dataset further support the effectiveness of our approach. Compared to Debate and Self-Consistency, our approach achieves comparable or superior performance while significantly reducing the average number of agent calls. Moreover, our framework demonstrates strong robustness across different model families and parameter sizes, confirming its applicability across a wide range of architectures. These findings highlight that selectively invoking debate, rather than applying it uniformly or relying solely on single agent responses, offers a more principled balance between accuracy and efficiency. Interestingly, MAD exhibits lower accuracy compared to other debate-based methods. Consistent with findings from Wang et al. (2024), we attribute this to its inherent tendency toward contradictory reasoning. While constructive disagreement encourages divergent insights, it also intensifies erroneous reasoning, deteriorating the quality of final responses.

LLM Agent	Shift	MAD	Debate	Ours
GPT-4o-mini	✓→✗	70.59	50.00	33.59
	✗→✓	29.41	50.00	66.41
LLaMA-3.3 70B	✓→✗	48.91	60.09	12.57
	✗→✓	51.09	39.91	87.43
Qwen-2.5 72B	✓→✗	63.87	50.00	39.91
	✗→✓	36.13	50.00	60.09
Mix	✓→✗	70.85	30.81	47.35
	✗→✓	29.15	69.19	52.65

Table 3: Proportions of correct and incorrect response changes before and after debate across multiagent collaboration methods. We denote a correct answer by ✓and an incorrect answer by ✗.

Mixed-Model Configuration Under the mixed-model configuration, external feedback originates from models trained on varying datasets and scales, which facilitates the emergence of diverse and complementary reasoning behaviors. As shown in Figure 3, DOWN consistently strong baselines, achieving higher accuracy with significantly fewer agent calls across both MUSR and StrategyQA. DOWN surpasses the Debate baseline in performance, with a markedly reduced computational burden. These results demonstrate that our conditional debate remains effective beyond homogeneous setups, adapting seamlessly to mixed-model collaboration. This highlights the robustness and scalability of our debate system in diverse reasoning environments.

Comparison of Final Answer Generation Strategies Voting-based selection and judge-based evaluation are two strategies for consolidating multiagent debate responses into a final decision. Experimental results indicate that both approaches yield similar response patterns, with marginal differences depending on the specific setting. However, the judge-based method requires an additional agent call, making it slightly less efficient. Given